CHANGYUE (WILLIAM) ZHAO

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EDUCATION

Duke University, Durham, NC *08/2025 – Present*
Master of Biostatistics, Health AI Track
- GPA: 4.0/4.0
- *Selected Coursework: Statistical Methods for Learning and Discovery, Deep Learning for Health Data, Natural Language Processing, High-Dimensional Statistics and Machine Learning, Real Analysis (Instructor: Michael C. Reed, ranked 2nd), Statistical Inference (Ph.D. Level)*

University of Minnesota, Twin Cities, Minneapolis, MN *09/2023 – 05/2025*
Bachelor of Science in Statistical Science, Minor in Computer Science
- GPA: 4.0/4.0
- *Selected Coursework: Mathematical Foundation (Introduction to Mathematical Analysis), Regression and Correlation Data, Statistical Computing in R, Theory of Statistics, Statistical Machine Learning, Analysis of Categorical Data, Applied Multivariate Methods, Discrete Mathematics, Advanced Programming Principles, Algorithms and Data Structures, Applied Machine Learning, Data Mining (Graduate Level)*
- Graduated with High Distinction.
- Received the Buehler Memorial Undergraduate Award as one of eight nominees.
- Named to the CLA Dean's List in all four semesters (4/4).

Shandong University of Finance and Economics, Jinan, China *09/2021 – 06/2025*
Bachelor of Economics in Finance
- GPA: 92.86/100 (WES GPA: 3.89/4.0)
- *Selected Coursework: Advanced Mathematics (Calculus, Introduction to Ordinary Differential Equations), Linear Algebra, Probability Theory and Statistics, Applied Statistics, Python*
- Received the SDUFE First-Prize Scholarship in 2022 and 2023.

RESEARCH INTERESTS

Applications of foundation models and LLM-based agents for biomedical analysis
Deep learning methods for biomedical data analysis and multimodal integration

PUBLICATIONS

- **Zhao, C.**, and Ji, Z. (2026). Visual LLM-guided consensus spatial domain detection with L-STAR. *Nature Communications*. doi:10.1038/s41467-026-78923-w.
- **Zhao, C.**, and Ji, Z. TRACE: Trajectory Reasoning via AI-guided Cross-view Evidence. *Manuscript in preparation.*

RESEARCH EXPERIENCE

TRACE: LLM-guided Trajectory Inference *01/2026 – Present*
Department of Biostatistics and Bioinformatics, Duke University, Durham, NC
- Develop TRACE, an LLM-guided trajectory inference agent that achieves high agreement with reference trajectories across evaluated scRNA-seq, scATAC-seq, and spatial transcriptomics datasets while automatically identifying root states from molecular evidence.
- Build `trace`, an end-to-end Python package using `LangGraph` to coordinate R/Seurat analysis and LLM reasoning. Use Seurat for normalization, dimensionality reduction, clustering, and marker-gene analysis, and Signac for scATAC-seq preprocessing and gene-activity estimation.
- Design structured prompts and targeted molecular-evidence queries to integrate UMAP and t-SNE views for cell-type annotation, lineage reconstruction, and root identification.
- Develop spatial trajectory inference combining molecular evidence and spatial neighborhood graphs to distinguish repeated local processes. Implement graph-based pseudotime estimation with local root cells, retaining ambiguous assignments and partially resolved trajectories.
- Design `pPOS` to jointly evaluate lineage membership and within-lineage ordering. Implement blockwise pair counting and inclusion-exclusion to avoid double-counting cell pairs shared across branches.

- Investigate discrepancies in `dyneval`-based `F1branches` evaluation. Reimplement the metric according to its published definition over the full cell set, with diagnostics for cell coverage and branch recovery.
- Implement comparative evaluation of TRACE, Monocle 3, Slingshot, and TSCAN using branch membership and within-lineage pseudotime ordering.
- **Supervisor:** Prof. Zhicheng Ji, Duke University

Med2State *11/2025 – Present*

School of Medicine, Duke University, Durham, NC

- Develop Med2State, a deep learning model in Python and PyTorch that learns structured patient-state representations from incomplete multimodal records and longitudinal clinical history, with separate disease, procedure, and treatment channels.
- Build modular encoders for demographics, measurements, structured EHR events, text, and images. Explore semantic cross-attention, low-rank fusion, and temporal models for irregular visits.
- Develop learning objectives and clinical concept readouts for incomplete longitudinal records.
- **Supervisor:** Prof. Anru Zhang, Duke University

L-STAR: LLM-guided Spatial Domain Detection *Sep 2025 – Oct 2026*

Department of Biostatistics and Bioinformatics, Duke University, Durham, NC

- Developed L-STAR, an LLM-guided spatial domain detection framework that achieved the highest average normalized ARI across evaluation datasets, outperforming individual methods and fixed-method consensus.
- Designed structured prompts and implemented Python workflows to compare candidate domain maps using tissue context and optional histology.
- Implemented repeated pairwise comparisons and win-rate ranking for method selection. Combined selected domain assignments through co-assignment matrices and consensus clustering in R.
- Processed spatial expression data with Seurat and SpatialExperiment and built marker-guided domain annotation. Distinguished tissue regions from cell types by sampling level, retaining Unknown for insufficient evidence. Evaluated agreement using BioLORD-2023 embeddings and within-dataset permutation tests.
- Benchmarked L-STAR across six spatial transcriptomics datasets against individual methods and fixed-method consensus. Assessed sensitivity to ensemble size, LLM choice, color palettes, and presentation order, and quantified selected-method redundancy.
- **Open-source software:** L-STAR **Supervisor:** Prof. Zhicheng Ji, Duke University

PaperEval *03/2025 – 09/2025*

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Built resumable OpenReview pipelines using `Selenium` and asynchronous requests, with structured `JSON` storage for papers, reviews, scores, and responses. Extracted author contact information from PDFs with `PyMuPDF` to enrich submission metadata.
- Prepared training data and fine-tuned GPT-4o for paper acceptance prediction. Evaluated predictions using progressively richer inputs comprising paper metadata, reviewer scores, and review text.
- Developed LLM-based agents to assess figures and tables in the context of full manuscripts and suggest improvements to clarity, readability, and presentation.
- Fine-tuned GPT-4o mini as an LLM-based grader to score and compare author responses. Constructed proxy scoring targets from acceptance levels using bounded Gaussian noise.
- Trained a reply-generation model and used the fine-tuned grader to compare generated responses with author replies from accepted and rejected submissions.
- **Open-source software:** PaperEval **Supervisor:** Prof. Jie Ding, University of Minnesota

MUDAC (Midwest Undergraduate Data Analysis Competition) 2025 *03/2025 – 04/2025*

School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Engineered features from sparse, high-dimensional numerical and text data for mentoring-match prediction, including embeddings, sentiment scores, and pairwise similarities.
- Modeled match duration and closure using Cox proportional hazards regression and a causal Bayesian network.
- Designed a mixture-of-experts model with regression, MLP, and Transformer experts for different feature types and a learned gating network. Compared it with random forest and XGBoost baselines.
- Placed 7th in Innovation & Creativity among approximately 200 teams at MUDAC 2025.
- **Supervisors:** Barbara Kuzmak and Robert Oliver VandenBurg, University of Minnesota

UMN & Hennepin County Public Health Equity Data Challenge *10/2024 – 11/2024*
School of Public Health, University of Minnesota, Twin Cities, Minneapolis, MN

- Investigated inequities in access to mental-health resources using Hennepin County questionnaire data. Applied multiple regression, ridge regression, decision trees, and causal inference methods.
- Developed a `tkinter` interface for mental-health prediction using `AdaBoost` in `scikit-learn`.
- Co-authored the analysis report and poster and placed in the top 25% of the undergraduate division.
- **Supervisors:** Marta Shore and Prof. Julian Wolfson, University of Minnesota

Overlapping Group LASSO Regression R Package Implementation *09/2024 – 05/2025*
School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Compared `overlapping_LeastR` from the MATLAB `SLEP` library with `oglasso` in the R package `VennLasso` under matched problem sizes and convergence tolerances.
- Designed a simulation study with 1,000 large sparse-matrix datasets and overlapping groups. Compared implementation runtimes using t-tests.
- Implemented the optimized overlapping-group LASSO algorithm in C for integration with R.
- **Supervisor:** Prof. Tianxi Li, University of Minnesota

PROFESSIONAL EXPERIENCE

Teaching Assistant: BIOSTAT 830 *08/2026 – Present*
Department of Biostatistics and Bioinformatics, Duke University, Durham, NC

- Help develop and refine BIOSTAT 830, a new course first offered in Fall 2026 to bridge undergraduate statistical theory and PhD-level inference. Quickly adapt lecture materials and proof-based assignment solutions while building course resources from the ground up.
- Hold weekly office hours and grade assignments for a class of 22 students in statistical inference, asymptotic theory, and decision theory. Discuss common errors with the instructor to guide future lectures.
- Prepare and verify proof-based LaTeX solutions, format and proofread slides, and manage materials in Canvas. Standardize notation and convert formulas into editable PowerPoint equations.
- **Supervisor:** Dr. Lin (Lynn) Lin, Duke University

Undergraduate Teaching Assistant: STAT 3011 *09/2024 – 05/2025*
School of Statistics, University of Minnesota, Twin Cities, Minneapolis, MN

- Taught weekly 50-minute sections for about 30 students on probability, sampling, hypothesis testing, regression, and statistical programming in R.
- Prepared instructional slides and managed course materials in Canvas for the teaching team. Held office hours and graded weekly assignments and exam responses through Gradescope.
- English-language teaching qualification: ELP 1, the highest rating on the Spoken English Test for Teaching Assistants (equivalent to TOEFL iBT Speaking 27–30/30 or IELTS Speaking 8.5–9.0/9).
- **Course materials:** GitHub repository
- **Supervisors:** Yuyoung Park, Aritra Banerjee, Kazeem Adepoju, University of Minnesota

OTHER INFORMATION

Programming	Python, R, C/C++, Java, OCaml, Swift, MATLAB, SQL
Machine Learning	`PyTorch`, `TensorFlow`, `scikit-learn`, Hugging Face, NumPy, pandas
Bioinformatics	Seurat, Signac, Scanpy, scDesign3, Monocle 3, Slingshot, TSCAN
Tools	Git, SQLite, Selenium, `LangGraph`, LaTeX, Markdown, SPSS
Hobbies	Photography, Snowboarding, Car Racing, Cycling, Fitness Exercise